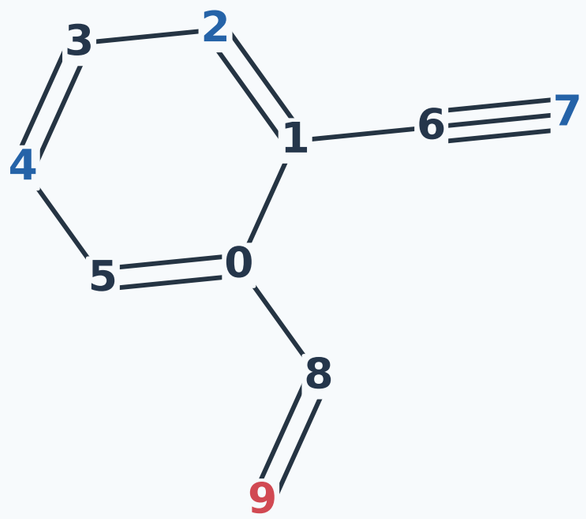
1 A numbered structure becomes two lists

A graph is a list of atoms plus a list of connections. The drawing coordinates only help us see it.

Numbered chemical structure



Atom list

ID	element	neighbors
0	C	3
1	C	3
2	N	2
3	C	2
4	N	2
5	C	2
6	C	2
7	N	1
8	C	2
9	O	1

Connection list

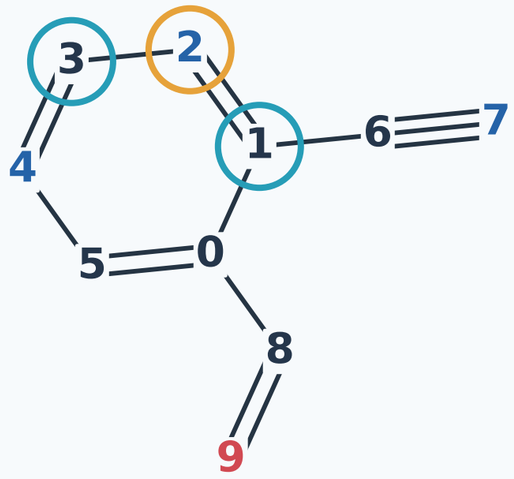
atoms	chemical bond	π coupling
0–1	single	-1.0
1–2	double	-1.0
2–3	single	-1.0
3–4	double	-1.0
4–5	single	-1.0
5–0	double	-1.0
6–7	triple	-1.0
1–6	single	-0.9
8–9	double	-1.0
0–8	single	-0.9

The model gets atom and connection data. It does not get the x–y positions in this picture.

2 Each atom and connection becomes numbers

Example: the highlighted ring nitrogen (atom 2) and its two direct neighbors.

Gold: atom 2 • teal: neighbors 1 and 3



Atom 2: category flags

Element is N:

C N O S F Cl

0 1 0 0 0 0

π class is N_pyridine:

0 1 0 0 0 0 0 0

Atom 2: four numerical values

formal charge / 2 = 0

neighbor count / 4 = 0.5

π electrons / 2 = 0.5

Hückel site value / 2 = -0.25

Together: 6 + 8 + 4 = 18 numbers

Atom 2's two connections

atom pair	bond order	π coupling
1–2	double = 2	-1.0
2–3	single = 1	-1.0

Repeat for the whole molecule

10 active atoms → 10 rows × 18 numbers

10 bonds → two connection grids:

bond orders and π couplings

Arrays are padded to 41 atoms;

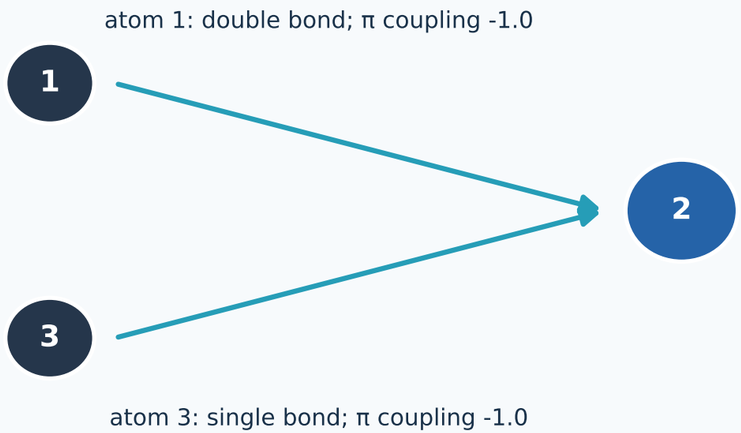
a mask hides the unused rows.

A category flag is a 1 for the matching category and 0 for the others; these are inputs, not predictions.

3 The model shares information along those connections

One update around atom 2, then the same operation across every atom and four rounds.

A local update at atom 2



What the model learns to do

Read atom 2's own 18-number description.

Read the descriptions of its neighbors 1 and 3.

Treat single and double bonds differently.

Use the π-coupling values as edge weights.

Write an updated description for atom 2.

1 Translate

18 input numbers per atom become a 96-number working description

2 Exchange

Every atom combines its own information with connected neighbors

3 Repeat ×4

After four rounds, an atom can reflect information from up to four bonds away

4 Predict

Summarize all atoms into one molecule description; predict the HOMO–LUMO gap

The π-only graph follows the same idea using Hückel couplings, three numbers per site, and six rounds. Neither graph model uses transformer attention.